THE NOVA RESEARCH CURRICULUM

Answer key · Grade 8 · Term 1

For teachers only. Do not share with students.

Teacher answer key

English

Question bank

#	Outcome	Question	Answer
1	EN-8.1	The main purpose of a tourism leaflet is to ...	entertain / **persuade** / instruct / warn
2	EN-8.1	"The best city in the world" is a fact. True or false.	**False** (it is an opinion)
3	EN-8.1	Which makes a source more reliable?	no author / **evidence and a named expert** / lots of capital letters / one opinion only
4	EN-8.1	"Loaded" words are words that ...	are very long / **carry strong feelings to sway the reader** / are spelled wrongly / are technical
5	EN-8.1	Write one sentence comparing the purposes of Texts A and B.	Model: Whereas Wordsworth's poem shares a personal feeling about nature, the leaflet aims to persuade tourists to visit Swat.
6	EN-8.2	In "Although it rained, we played," the subordinate clause is ...	we played / **although it rained** / it rained we / the whole sentence
7	EN-8.2	Choose the relative pronoun: The girl ___ won the prize is my cousin.	which / **who** / whose / where
8	EN-8.2	Combine: "I stayed home. I was ill."	Model: I stayed home because I was ill.

Urdu

Question bank

#	Outcome	Question	Answer
1	UR-8.1	علامہ اقبال کہاں پیدا ہوئے؟	لاہور / **سیالکوٹ** / علی گڑھ / کراچی
2	UR-8.1	«ترانۂ ملی» کا بنیادی پیغام ہے ...	سیر و تفریح / **ملتِ اسلامیہ کی وحدت** / موسم / تجارت
3	UR-8.1	«مسدس» کس کی مشہور نظم ہے؟	غالب / اقبال / **حالی** / میر

#	Outcome	Question	Answer
4	UR-8.1	سرسید احمد خان نے مسلمانوں کی تعلیم کے لیے کام کیا۔ True or false.	**True**
5	UR-8.3	جملے کے آخر میں کون سی علامت لگتی ہے؟	سکتہ / **ختمہ** / واوین / رابطہ
6	UR-8.3	« » کو کیا کہتے ہیں؟	سکتہ / وقفہ / **واوین** / ندائیہ
7	UR-8.3	سوالیہ جملے کے آخر میں ؟ لگتا ہے۔ True or false.	**True**

Mathematics

Question bank

#	Outcome	Question	Answer
1	MA-8.1	Which is irrational?	0.5 / 1/3 / **√2** / √9
2	MA-8.1	0.333... (recurring) is a rational number. True or false.	**True**
3	MA-8.1	$2^3 \times 2^4 = ?$	2^{12} / $\mathbf{2^7}$ / 4^7 / 2^1
4	MA-8.1	$5^0 = ?$	0 / 5 / **1** / undefined
5	MA-8.1	$3^{-2} = ?$	−9 / −6 / **1/9** / 9
6	MA-8.1	3,476 to 2 significant figures?	3,400 / **3,500** / 35 / 3,480
7	MA-8.7	Marked price Rs 2,000 with 15% discount. Sale price?	Rs 1,850 / **Rs 1,700** / Rs 300 / Rs 2,300
8	MA-8.7	Cost Rs 800, markup 25%. Selling price?	Rs 825 / Rs 900 / **Rs 1,000** / Rs 1,200
9	MA-8.7	A and B invest Rs 30,000 and Rs 20,000. Profit Rs 10,000. A's share?	Rs 5,000 / **Rs 6,000** / Rs 4,000 / Rs 7,500
10	MA-8.7	At an example rate of 1 US dollar = Rs 280, how many rupees is 50 dollars?	Rs 5,600 / **Rs 14,000** / Rs 1,400 / Rs 330

Physics

Question bank

#	Outcome	Question	Answer
1	PH-8.1	The angle of incidence is 35°. The angle of reflection is ...	55° / **35°** / 70° / 90°
2	PH-8.1	Angles of incidence and reflection are measured from the ...	mirror / **normal** / floor / light source
3	PH-8.1	Vehicle side mirrors are ...	plane / concave / **convex** / glass blocks

#	Outcome	Question	Answer
4	PH-8.1	Light going from air into water bends ... the normal.	away from / **towards** / along / it does not bend
5	PH-8.1	A magnifying glass uses a ... lens.	concave / **convex** / flat / coloured
6	PH-8.1	A green leaf looks green because it ... green light.	absorbs / **reflects** / makes / blocks
7	PH-8.1	A plane-mirror image is upright and the same size. True or false.	**True**
8	PH-8.1	Why does a swimming pool look shallower than it is?	Model: light from the bottom refracts away from the normal as it leaves the water, so the bottom seems higher.

Chemistry

Question bank

#	Outcome	Question	Answer
1	CH-8.1	Elements in the periodic table are arranged by ...	name / colour / **atomic number** / discovery date
2	CH-8.1	Elements in the same group have the same number of ...	shells / neutrons / **outer electrons** / protons
3	CH-8.1	Which is a noble gas?	oxygen / **neon** / chlorine / sodium
4	CH-8.1	Noble gases are unreactive because ...	they are heavy / **their outer shell is full** / they are metals / they are liquids
5	CH-8.1	The symbol Na stands for ...	nitrogen / neon / **sodium** / nickel
6	CH-8.1	Metals are good conductors of electricity. True or false.	**True**
7	CH-8.1	Which non-metal conducts electricity?	sulfur / oxygen / **graphite (carbon)** / phosphorus
8	CH-8.1	Give two properties of metals.	Model: shiny and good conductors of heat and electricity (also: malleable, ductile).

Biology

Question bank

#	Outcome	Question	Answer
1	BI-8.1	Height is an example of ... variation.	discontinuous / **continuous** / no / chemical
2	BI-8.1	Blood group is an example of ... variation.	**discontinuous** / continuous / environmental only / none

#	Outcome	Question	Answer
3	BI-8.1	How many chromosomes are in a human body cell?	23 / **46** / 48 / 92
4	BI-8.1	A gene is a section of …	protein / **DNA** / cytoplasm / cell wall
5	BI-8.1	Mitosis produces …	four different cells / **two identical cells** / sex cells only / no cells
6	BI-8.1	Meiosis makes cells with half the number of chromosomes. True or false.	**True**
7	BI-8.1	Give one adaptation of a camel and explain it.	Model: wide feet stop it sinking into sand (also: stores fat in its hump, long eyelashes keep sand out).
8	BI-8.1	DNA has the shape of a …	single line / circle / **double helix** / square

History, Geography and Pakistan Studies

Question bank

#	Outcome	Question	Answer
1	HI-8.1	The War of Independence began in …	1757 / 1849 / **1857** / 1906
2	HI-8.1	Punjab was annexed by the British in …	1757 / 1843 / **1849** / 1858
3	HI-8.1	The last Mughal emperor, Bahadur Shah Zafar, was exiled to …	London / Kabul / **Rangoon** / Lahore
4	HI-8.1	After 1857 the British Crown took over from the East India Company. True or false.	**True**
5	HI-8.1	Give two causes of the War of 1857.	Model: annexation of states and the greased cartridges (also: heavy taxes, religious fears).
6	GE-8.1	Moving from a village to a city is called …	emigration / **rural-to-urban migration** / tourism / commuting
7	GE-8.1	Better jobs in a city are a … factor.	push / **pull** / natural / fixed
8	PS-8.1	The head of government in Pakistan is the …	President / **Prime Minister** / Chief Justice / Governor
9	PS-8.1	Pakistan's Parliament has two houses: the National Assembly and the …	Cabinet / **Senate** / High Court / Provincial Assembly

Islamiyat and Ethics

Question bank

#	Outcome	Question	Answer
1	IS-8.1	Luqman's advice is in Surah …	Al-Hujurat / **Luqman** / Al-Furqan / Al-Mulk

#	Outcome	Question	Answer
2	IS-8.1	Luqman told his son that shirk is …	a small mistake / **a great wrong** / allowed sometimes / a good deed
3	IS-8.1	Allah will bring out even a deed the weight of a …	stone / **mustard seed** / mountain / feather
4	IS-8.1	Luqman told his son to lower his voice. True or false.	**True**
5	IS-8.1	If parents push you towards shirk, you should …	obey them / leave them forever / **not obey in that, but stay kind to them** / argue rudely
6	IS-8.1	Name three things Luqman told his son to do in 31:17.	Model: establish prayer, enjoin good and forbid evil, and be patient.
7	IS-8.1	Walking proudly is praised in Surah Luqman. True or false.	**False**
8	ET-8.1	Your digital footprint is …	your shoe size / **the trail of data you leave online** / a virus / a password

Computer Science and skills

Question bank

#	Outcome	Question	Answer
1	CS-8.1	A missing colon after `if x > 5` causes a … error.	logic / **syntax** / runtime / network
2	CS-8.1	A program that runs but gives the wrong answer has a … error.	syntax / **logic** / spelling / hardware
3	CS-8.1	`try` and `except` help a program handle bad input without crashing. True or false.	**True**
4	CS-8.1	Which module gives random numbers?	math / time / **random** / os
5	CS-8.1	Name three kinds of test input.	Model: normal input, edge (boundary) input and wrong input.
6	RO-8.1	The first step of the design cycle is to …	build / test / **identify the problem** / sell
7	FL-8.1	Rent for a shop is a … cost.	variable / **fixed** / hidden / zero
8	RP-8.1	Interviews need the person's permission. True or false.	**True**